Robotyx Systems · Drilling World

# Drilling World user guide

## Version v5.12.20

How to use the showroom, Rig Builder, frac, DD tracks, settings, Cool mode, EDM map, site sync and cloud sync and the WITSML dashboard.

Edition: 1 November 2026 (AEST)

Written in ASD-STE100 Simplified Technical English. Screenshots from the live site with made-up test data.

# Contents

About this guide

Rig showroom

Assets slider

Rig Builder

Frac and plug-and-perf

DD tracks and cabin steering exercise

Settings toggles

Cool mode (arcade showroom)

EDM import

Map overlay (Satellite and Street)

GL labels and TD markers

Phone and laptop site sync

WITSML dashboard and image logs

# About this guide

This guide tells you how to use the Drilling World simulator at `https://drilling.robotyx.systems`.

| | |
|---|---|
| **Version** | v5.12.20 (screen stamps `v5.12.9` to `v5.12.20`) |
| **Screenshots** | Taken from the live site (v5.12.9; start screen v5.12.20) on 1 Nov 2026 (AEST), laptop 1440 × 900 and phone 390 × 844 |
| **Test data** | All EDM screens use a made-up test file, "SYNTH FIELD (TEST)". It contains no customer data. |
| **Data tags** | `SIM` = simulator model. `IMPORT` = from your EDM file. Simulated values always show SIM. |

> Most values in the game are SIM teaching models. Do not use them for a real well design.

## Start screen

The start screen opens when you load the site. It has 4 large options. Each option opens a page with its tiles and a **Back** tile.

| Option | Tiles on its page |
|---|---|
| **Drill** | Continue (only when you have a saved game), New simulation, DD Training, DD Tracks, Rig showroom |
| **Plan** | New well, Import EDM, Saved sites (only when you have saved sites) |
| **Monitor** | WITSML dashboard, Monitor a well |
| **Operations** | Workover & intervention, Safety World |

The top bar has **Log in as player** and **? User guide**. The bottom bar has Sound, Rig builder, Sync sites and the version.

1. Tap an option, for example **Drill**.
2. Tap a tile on its page.
3. To go back, tap **Back**, or press Esc, or press B on a gamepad.

Keyboard: arrow keys (or WASD) move, Enter selects, Esc goes back. Gamepad: d-pad or stick moves, A selects, B goes back. Links such as `?wdash` still open a feature directly.

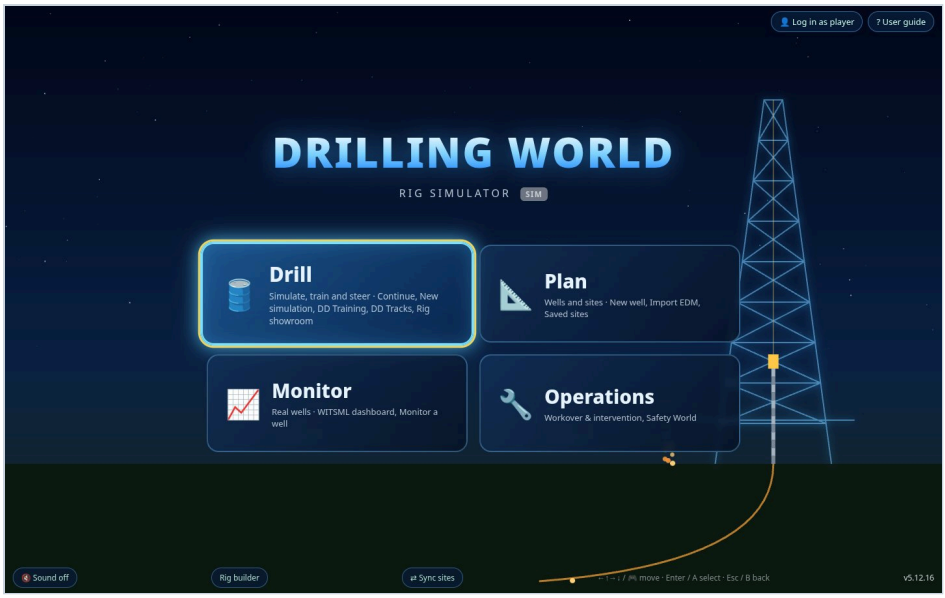



Figure 1a. Laptop (1440 × 900). Start screen with 4 options.




Figure 1b. Phone (390 × 844).

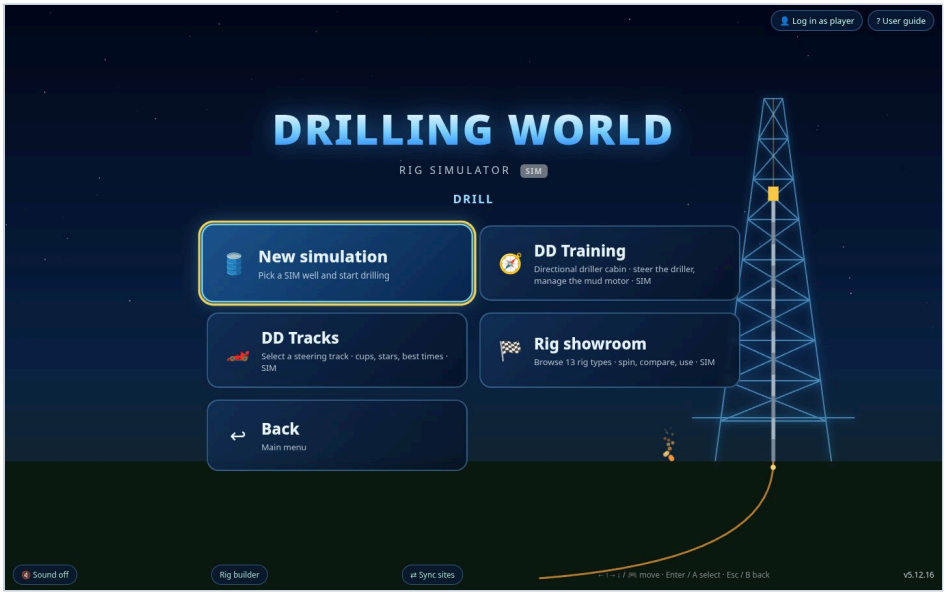



Figure 1c. Laptop. The Drill page.




Figure 1d. Phone. The Drill page.

# Rig showroom

The showroom shows one main land rig and 19 more rigs in one screen. Use it to look at rigs and their parts.

## Open the showroom

1. On the start screen, tap **Drill**, then **Rig showroom**.
2. Wait until the rig loads. The phone loads a lighter model.
3. Drag to turn the rig. Pinch or scroll to zoom.

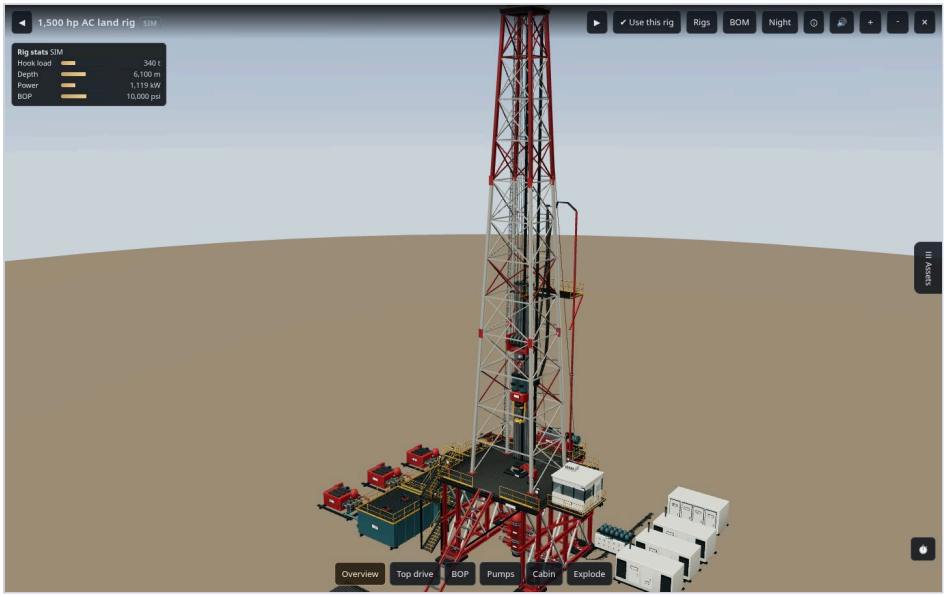



Figure 2a. Laptop (1440 × 900). Showroom with the main land rig.

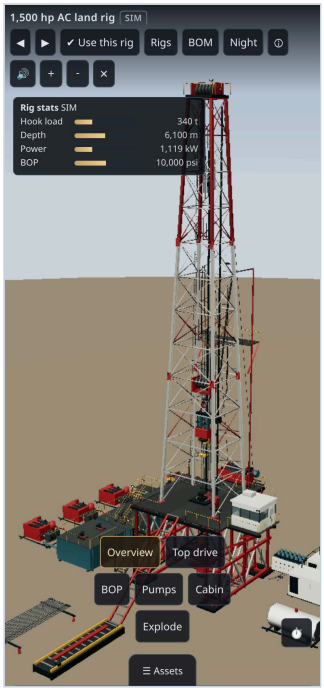



Figure 2b. Phone (390 × 844).

## Look at the rig

1. Tap **Overview**, **Top drive**, **BOP**, **Pumps** or **Cabin** for a close-up (keys 1 to 5).
2. Tap **Explode** (key X) to move the parts apart. Tap it again to close.
3. Tap **BOM** (key B) to see the parts list.
4. Tap **Night** (key N) for night lights.
5. Tap **Rigs** or press D and A to go to the next or previous rig.
6. Tap **Use this rig** to drill with the rig that you see.
7. Press Esc to go back one step.

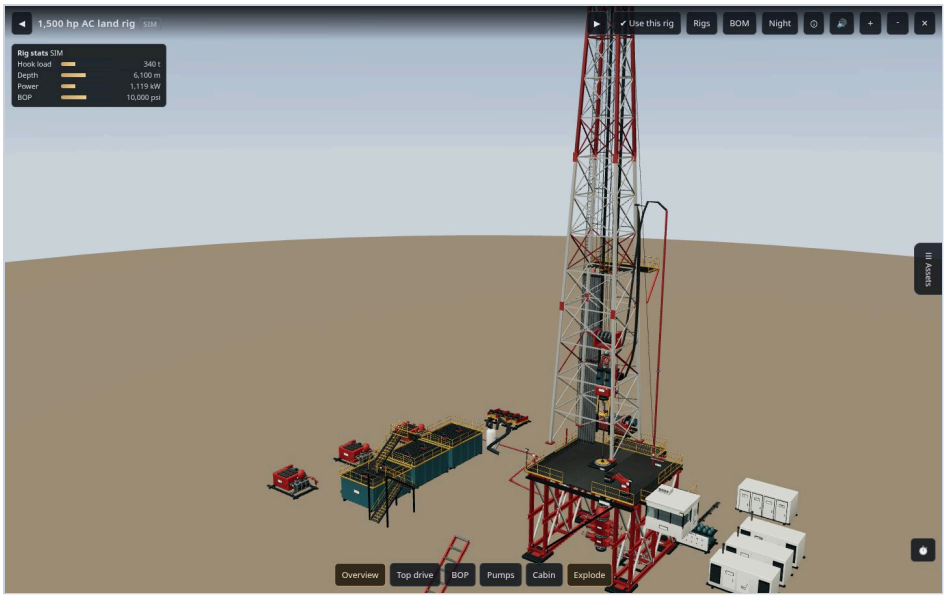



Figure 3a. Laptop (1440 × 900). Explode view.

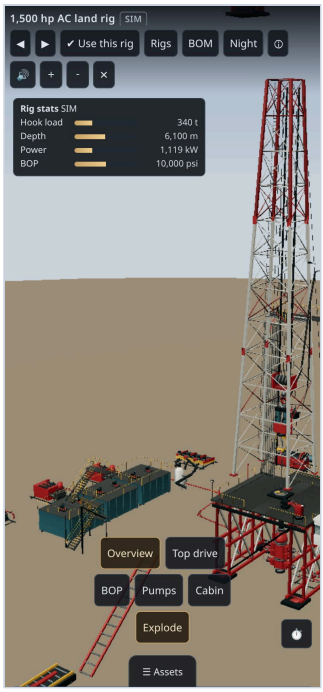



Figure 3b. Phone (390 × 844).

> The old Showroom A (classic) is off by default. To show it, use the Settings toggle (see Settings toggles).

# Assets slider

The Assets slider lists the parts of the rig that you see. Use it to find a part and zoom to it.

1. Open the showroom.
2. Tap **Assets** at the bottom of the screen. The slider opens.
3. Tap a row in the list. The camera zooms to that part.
4. Use the zoom buttons to zoom in and out.
5. Press Esc to close the slider.

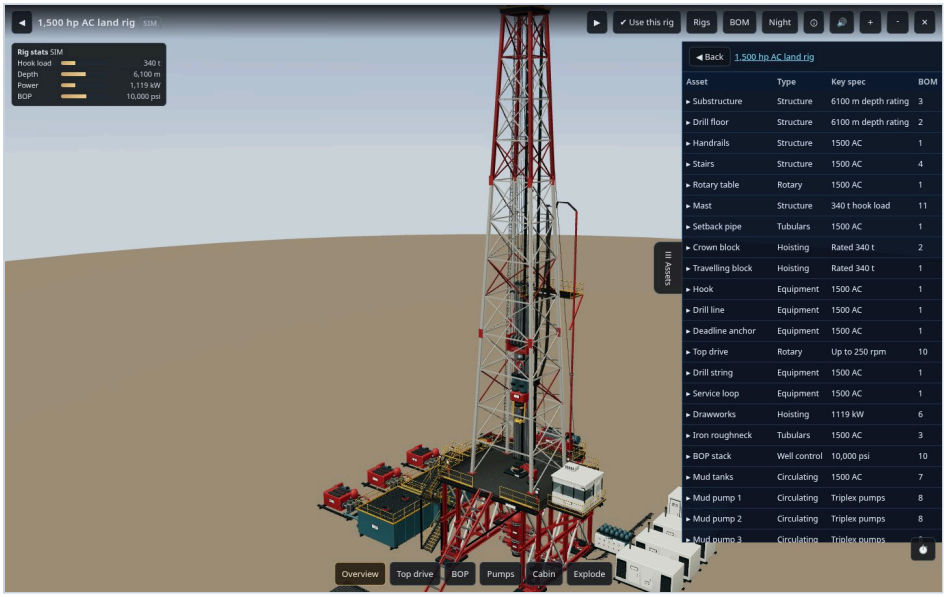



Figure 4a. Laptop (1440 × 900). Assets slider open with the part list.

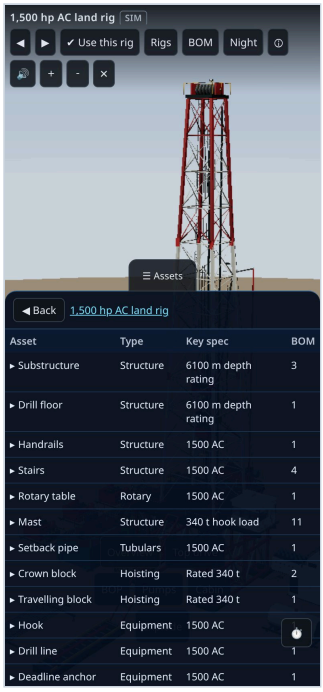



Figure 4b. Phone (390 × 844).

# Rig Builder

Rig Builder lets you build your own rig from parts. The specs and the BOM update when you change a part.

1. On the start screen, tap **Rig builder**.
2. In the left list, tap a slot (for example **Mast / derrick**).
3. Tap a part in the list below the slots.
4. Set the number of mud pumps and generators with the − and + buttons.
5. Type a rig name. Tap **Save**.
6. Tap **View in showroom** to see your rig in 3D.
7. Tap **Saved rigs** to open a rig that you saved.

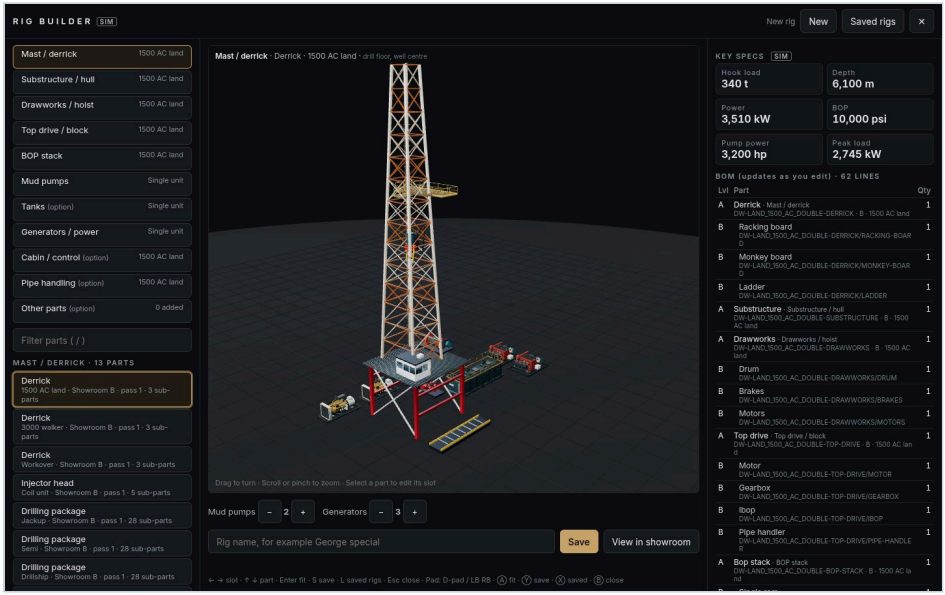



Figure 5a. Laptop (1440 × 900). Rig Builder with slots, specs and BOM. The 3D view was still loading part models when this shot was taken.

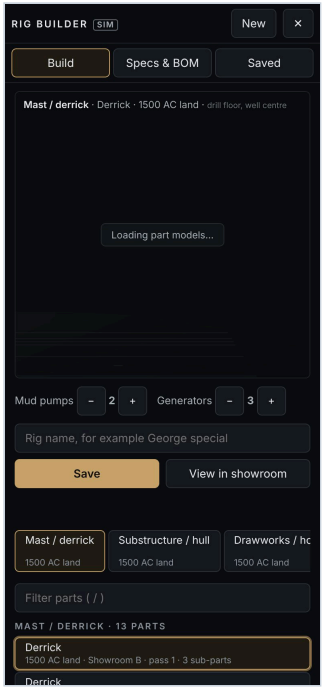



Figure 5b. Phone (390 × 844).

# Frac and plug-and-perf

The frac job shows a 3-stage plug-and-perf treatment. All values are SIM.

1. On the start screen, tap **Operations**, then **Workover & intervention**.
2. In **Sequence**, select **Hydraulic fracturing: plug-and-perf, 3 stages**.
3. Tap **Play**. Or tap **Step** to do one step at a time.
4. Read the HUD: stage, rate, treating pressure, proppant, fracture size and screenout risk.
5. Read the chart: pressure, rate and proppant against time.
6. Tap **Surface** to see the frac spread with the 6 pump trucks. Tap **Whole well** to see the stages downhole.

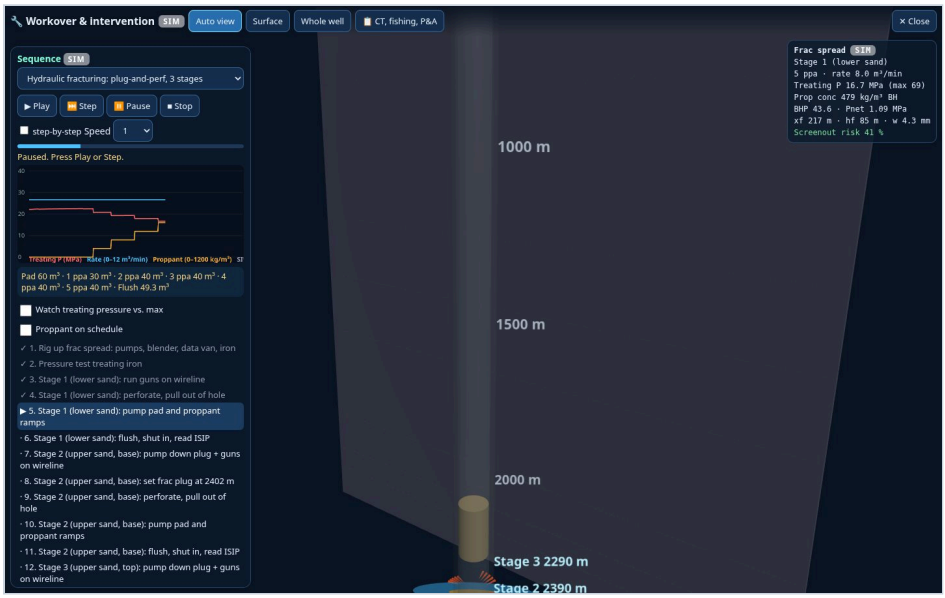



Figure 6a. Laptop (1440 × 900). Frac job at stage 1 (lower sand). HUD tag SIM.

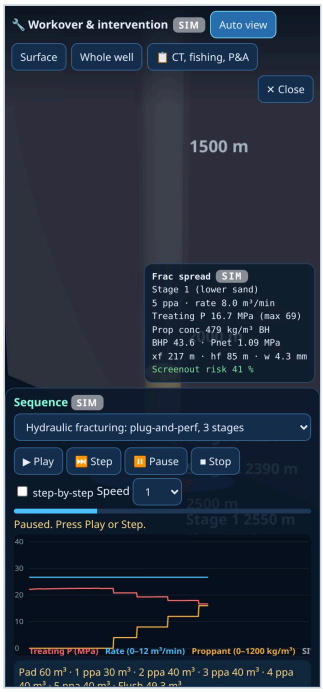



Figure 6b. Phone (390 × 844).

> To open the frac job directly, use `?workover=frac` at the end of the address.

# DD tracks and cabin steering exercise

DD Tracks is a steering exercise. You select a track, then you steer the well in the directional drilling cabin.

## Select a track

1. On the start screen, tap **Drill**, then **DD Tracks**.
2. Select a cup: **Rookie Cup**, **Pro Cup** or **Expert Cup**. Locked tracks show a lock.
3. Tap a track card. Use the arrow keys or the gamepad if you like.
4. Tap **START**. A countdown starts.

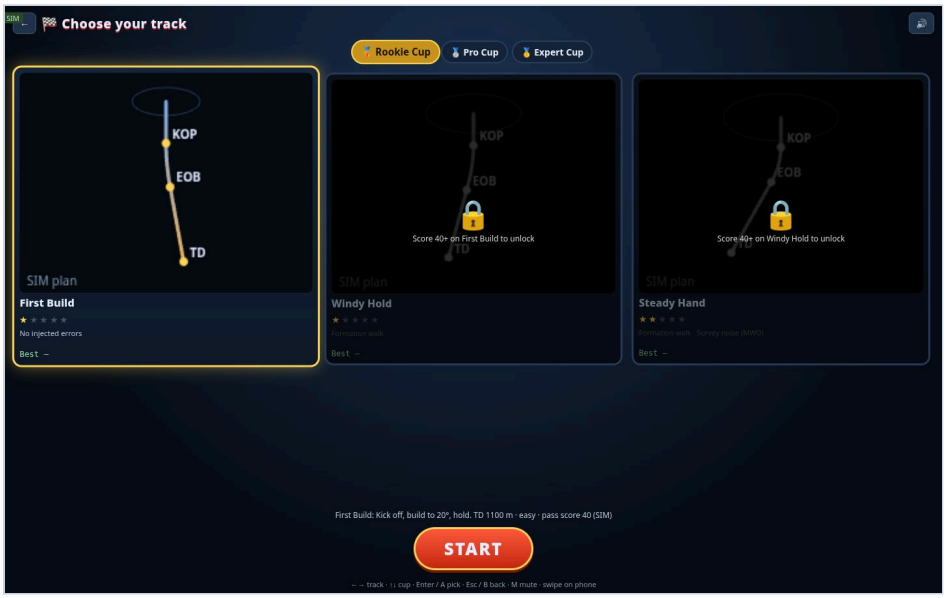



Figure 7a. Laptop (1440 × 900). Track select.

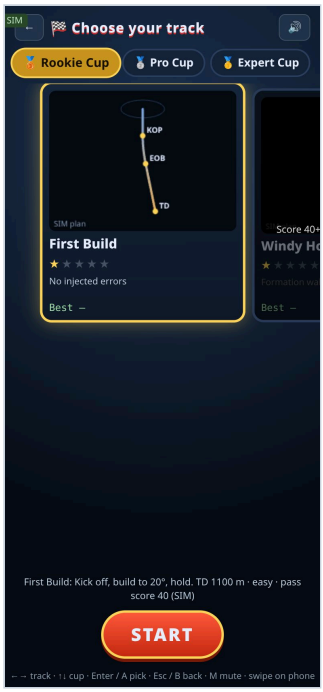



Figure 7b. Phone (390 × 844).

## Steer in the cabin

1. Read the instruction box. It tells you to slide or rotate.
2. Set the mode (**Slide** or **Rotate**), the toolface and the length.
3. Tap **Use instruction** or drill with your own values.
4. Read the survey table, the projection to bit and the travelling cylinder.
5. Keep the well near the plan. The score shows how well you steer.
6. Tap **Pause** or **Reset** when necessary. Tap **Tracks** to select a different track.

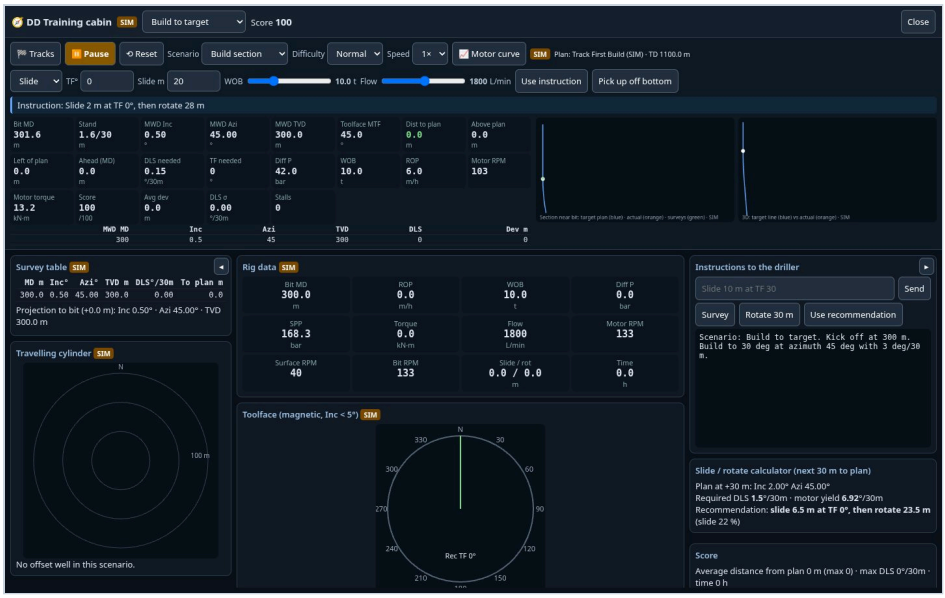



Figure 8a. Laptop (1440 × 900). DD cabin during a track run. All data SIM.

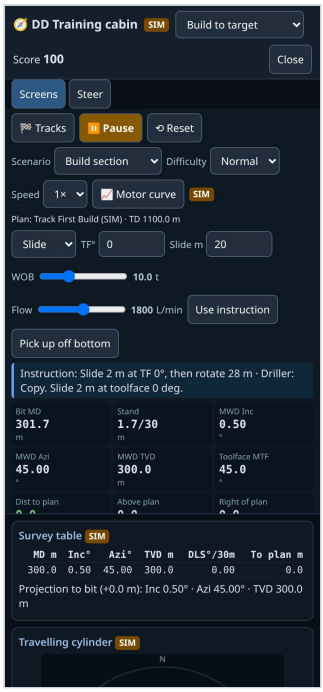



Figure 8b. Phone (390 × 844).

# Settings toggles

Open this guide from the game: tap **? User guide** in the top bar of the start screen, or **User guide (HTML / PDF)** in Settings. The PDF is at `/drill-sim-user-guide.pdf`.

Settings are in **Tools**. On the phone, tap **Tools** at the bottom of the screen.

1. Tap **Tools**. Go to **Settings**.
2. Set **Units** to metric or field.
3. Set **Camera**: Orbit, Walk floor or Follow bit.
4. Set **On-screen controls** to ON or OFF.
5. Tap **Controls guide** to see the keyboard, gamepad and screen controls.
6. Select **Show Showroom A (classic)** to show the old showroom. Clear it to hide it.
7. Use **Feed** settings only when you connect a WITSML feed. Without a feed, data stays SIM.




Figure 9a. Laptop (1440 × 900). Settings in the Tools panel.

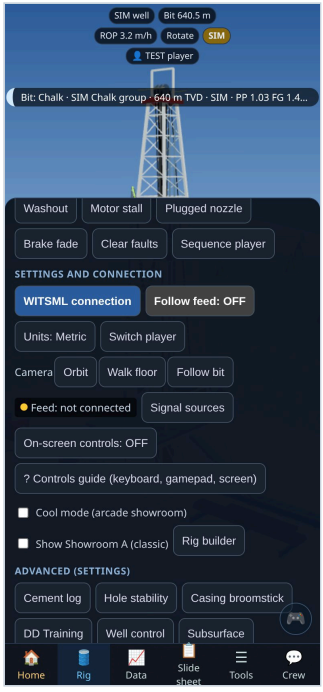



Figure 9b. Phone (390 × 844).

# Cool mode (arcade showroom)

Cool mode gives the Rig showroom the black arcade look and the kart-style rig select. Cool mode is OFF by default.

1. Tap **Tools**. Go to **Settings**.
2. Select **Cool mode (arcade showroom)**.
3. Open **Rig showroom**. The arcade rig select opens.
4. Use the ◀ and ▶ buttons or the arrow keys to change the rig. Tap **Select** to use the rig.
5. To go back to the standard showroom, clear **Cool mode (arcade showroom)**.

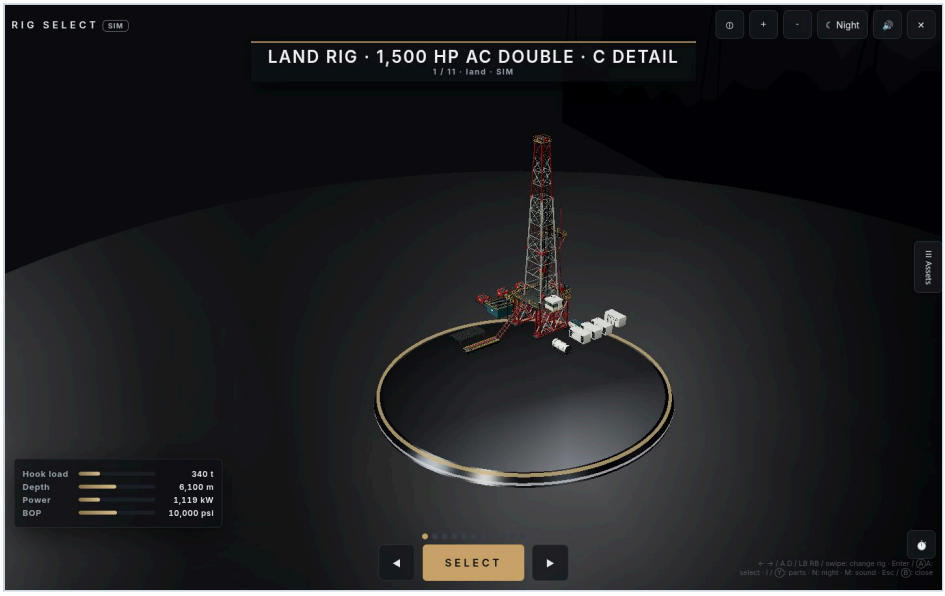



Figure 10a. Laptop (1440 × 900). Cool mode rig select. Rig values are SIM.

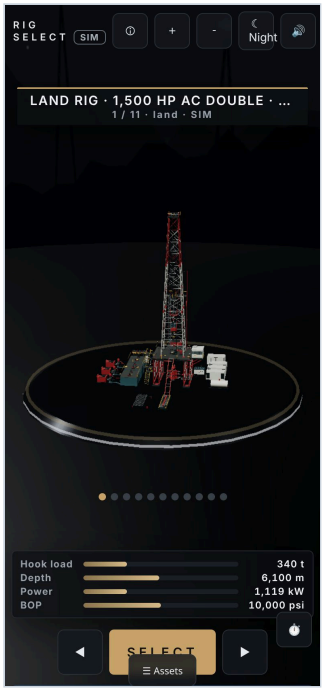



Figure 10b. Phone (390 × 844).

> The device keeps your Cool mode setting after a reload.

# EDM import

You can import an EDM export file (.xml or .zip). The file stays on your device. Values from the file show **IMPORT**.

1. On the start screen, tap **Plan**, then **Import EDM**.
2. Tap **Choose file** (or **Other file**). Select the EDM file.
3. Wait until the import is complete. The panel shows the number of sites, wells and wellbores.
4. Tap **Site 3D** to see the whole site in 3D.
5. Tap a well to see its well stats.

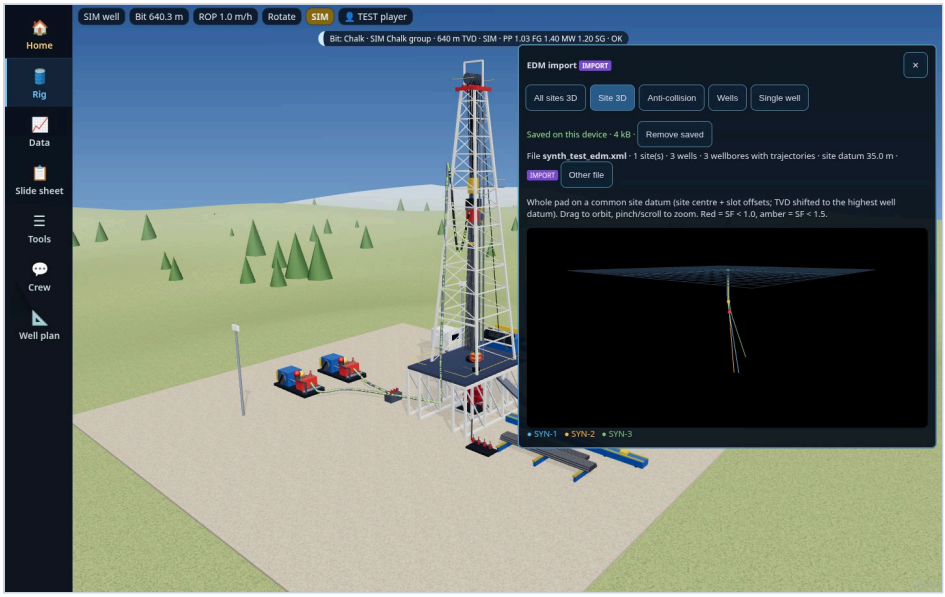



Figure 11a. Laptop (1440 × 900). EDM import with the test file (1 site, 3 wells, 3 wellbores).

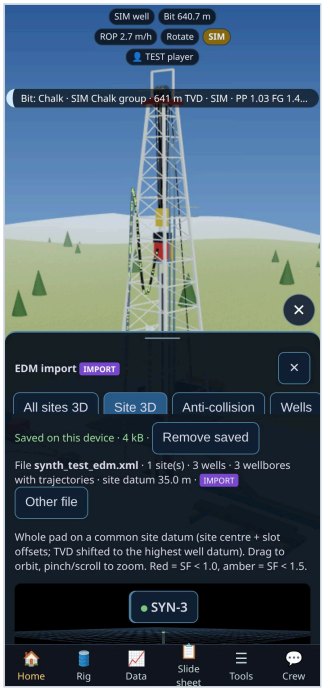



Figure 11b. Phone (390 × 844).

# Map overlay (Satellite and Street)

The site 3D view can show a map under the wells. The map uses the coordinate system (CRS) in the EDM file. Each wellhead is at its real position.

1. Open the site 3D view.
2. Tap the **Map** button. It changes from **Map: Off** to **Map: Satellite**.
3. Tap it again for **Map: Street**. Tap it again for **Map: Off**.
4. Read the attribution at the bottom of the map.

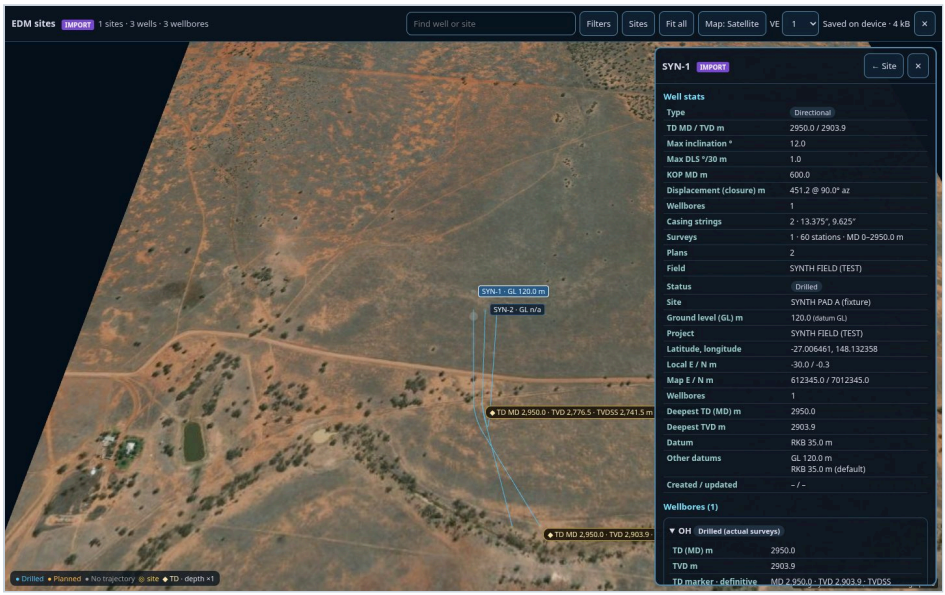



Figure 12a. Laptop (1440 × 900). Map: Satellite with the test site selected. Imagery © Esri, Maxar, Earthstar Geographics.




Figure 12b. Phone (390 × 844).

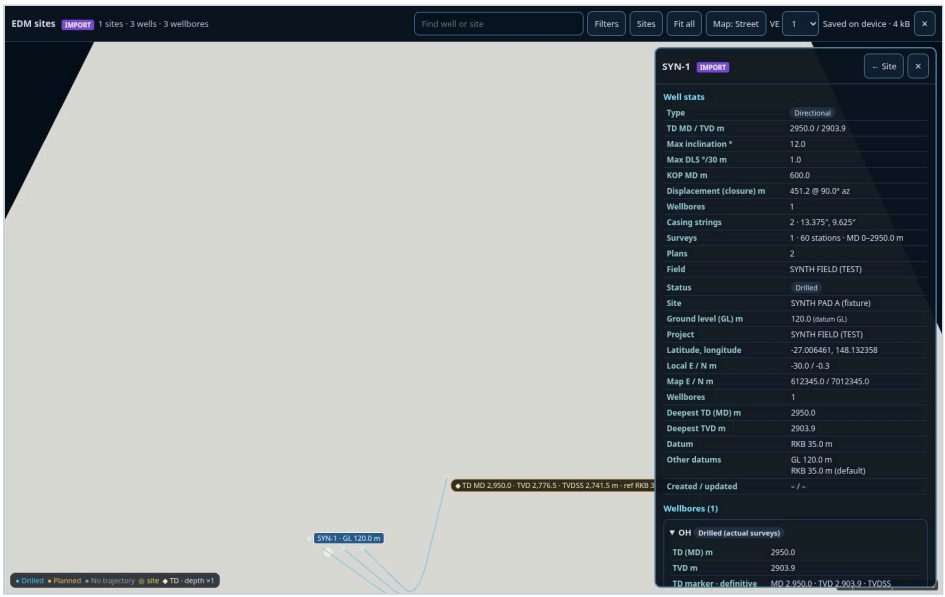



Figure 13a. Laptop (1440 × 900). Map: Street (© OpenStreetMap contributors).

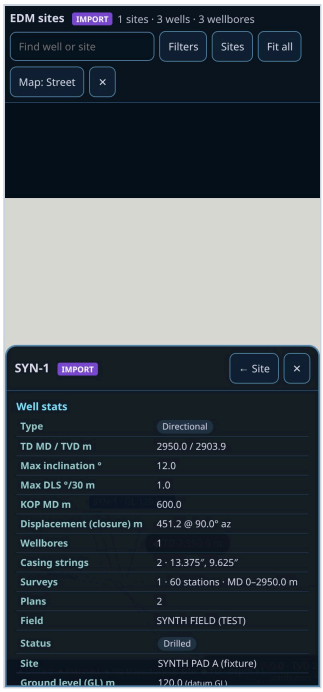



Figure 13b. Phone (390 × 844).

> The device keeps your map setting. Only the map tile requests leave the device. The EDM data does not leave the device.

> If the file has no CRS, the Map button does not show. A short note tells you why.

# GL labels and TD markers

Each wellhead label shows the ground level (GL) from the EDM file. Each wellbore and plan has a TD marker at its end.

1. Open the site 3D view. Tap a well or a site.
2. Read the wellhead label, for example `SYN-1 · GL 120.0 m`.
3. If the file has no ground level, the label shows `GL n/a`. The game does not guess a value.
4. Find the ◆ TD marker at the end of each wellbore.
5. On the laptop, read the full label: MD, TVD, TVDSS and the reference datum.
6. On the phone, tap the TD chip to see the full label.

| Device | Example TD label (test data) |
|---|---|
| Laptop | `◆ TD MD 2,950.0 · TVD 2,903.9 · TVDSS 2,868.9 m · ref RKB 35.0 m` |
| Phone | `◆ TD 2,950.0 m` (tap for the full label) |




Figure 14. Phone (390 × 844).

# Phone and laptop site sync

Site sync moves your saved EDM sites from one device to the other with a sync file. WITSML log-in data is never put in the file.

## Make the sync file (first device)

1. On the start screen, tap **⇄ Sync sites** in the bottom bar.
2. Tap **Make sync file**. Keep the file.

## Open the sync file (second device)

1. Send the file to the second device yourself.
2. On the start screen, tap **⇄ Sync sites** in the bottom bar.
3. Tap **Open sync file**. Select the file.
4. The saved sites show under **Plan** on the start screen.

The sync file also carries your custom rigs from Rig Builder. If the same site or rig is on both devices, the newer copy is kept.




Figure 15a. Laptop (1440 × 900). Sync saved sites panel with the sync file and Cloud sync parts.

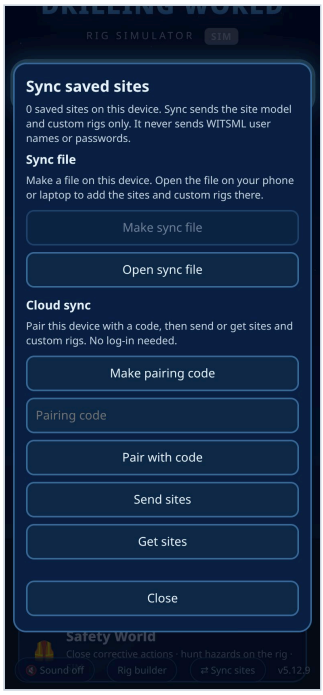



Figure 15b. Phone (390 × 844).

## Cloud sync (no sync file)

Cloud sync sends your saved sites and custom rigs through the Robotyx server. You do not need a log-in. Each device pairs with a 10-character code.

### Pair the first device

1. Tap **Sync**.
2. Tap **Make pairing code**. Write down the code.
3. Tap **Send sites**.

### Pair the second device

1. Tap **Sync**.
2. Type the code in **Pairing code**. Tap **Pair with code**.
3. Tap **Get sites**. The saved sites and custom rigs show.

> Cloud sync never sends WITSML user names or passwords. The server refuses data that looks like a credential. Keep the pairing code private.

> If cloud sync is not available, the panel shows "Cloud sync is off on this site". Use a sync file.

# WITSML dashboard and image logs

The WITSML dashboard shows drilling and formation logs in vertical tracks, side by side. It also shows image logs from array or sector curves. Without a store it shows `SIM` data. It shows **LIVE** only when it reads new data from a real WITSML store.

## Open the dashboard

1. On the start screen, tap **Monitor**, then 📊 **WITSML dashboard**.
2. Or, in Settings, tap 📊 **WITSML dashboard (logs + image logs)**.
3. Or, add `?wdash` at the end of the address.

## Read the logs

The default tracks are Drilling (ROP, WOB), Rotary (RPM, torque), Hydraulics (SPP, flow), Gamma (GR) and Resistivity. The track header shows the newest value and the scale. On the phone, swipe the tracks left and right.

1. Tap **Depth** or **Time** to change the index.
2. Tap **+** or **-** to zoom. You can also use the mouse wheel or drag the tracks.
3. Tap **Follow** to keep the newest data on the screen. When you drag, Follow stops. Tap **Follow** again to start it.

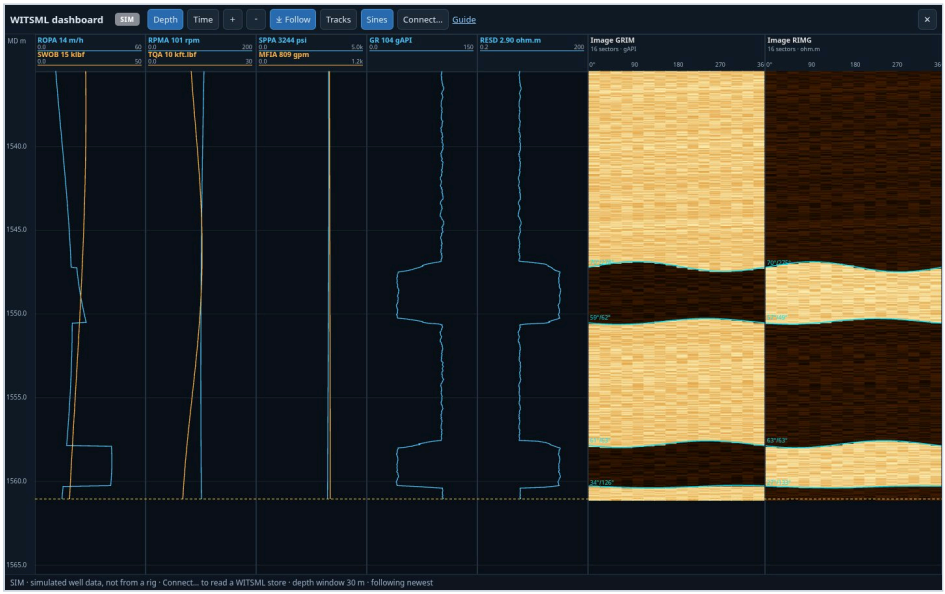



Figure 16a. Laptop (1440 × 900). Depth mode with SIM data.

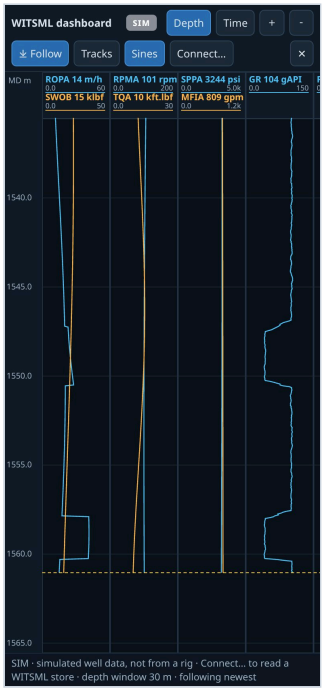



Figure 16b. Phone (390 × 844).

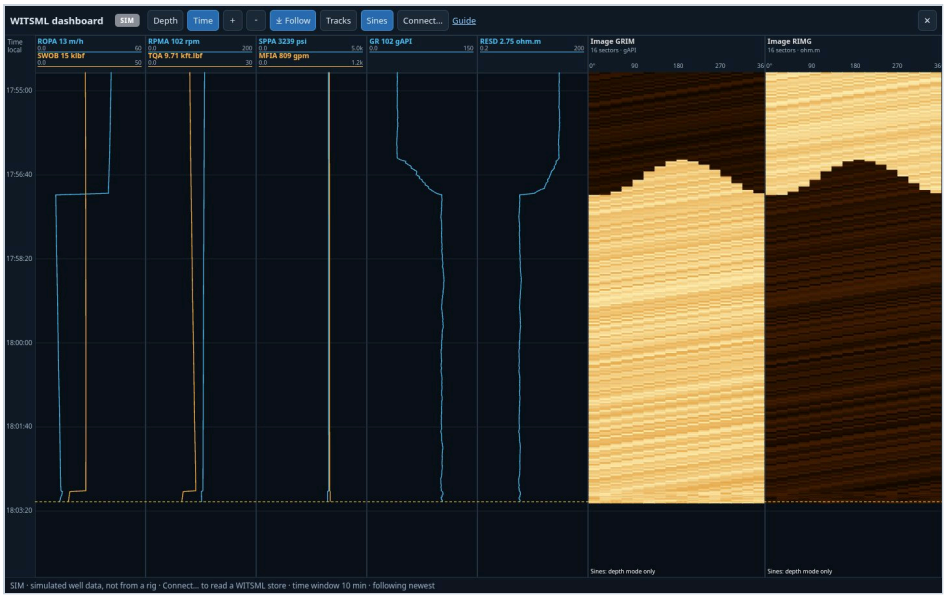



Figure 17a. Laptop (1440 × 900). Time mode with SIM data.

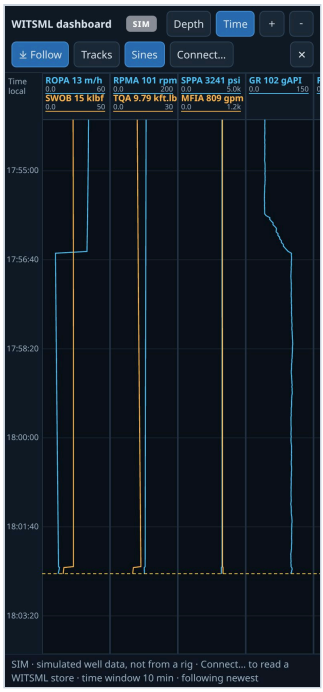



Figure 17b. Phone (390 × 844).

## Image logs

An image track unwraps the borehole wall from 0° to 360°. The dashboard finds array curves and sector curve families (for example `GRIM_01` to `GRIM_16`). Light colours are high values. Tap **Sines** to show the fitted sinusoids and the dip/azimuth labels (depth mode only).

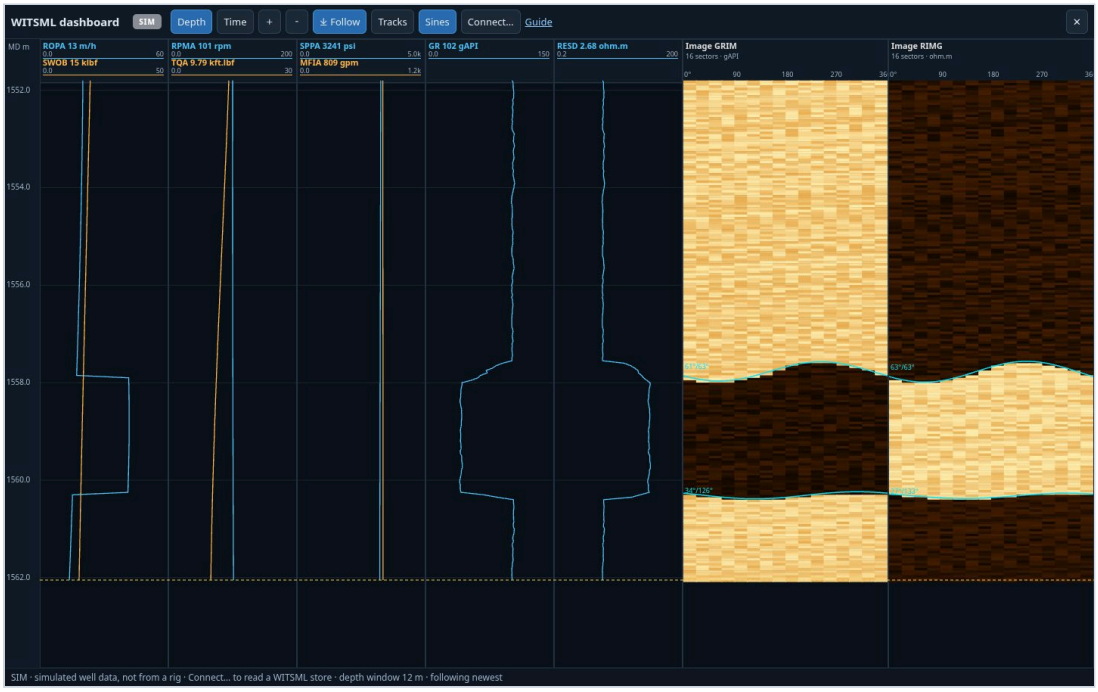



Figure 18a. Laptop (1440 × 900). Image tracks with sinusoids, zoomed in.

> If the log has no array or sector curves, the image track shows "No image (array / sector) curves". This is correct for logs without image data.

## Change tracks, curves and scales

1. Tap **Tracks**.
2. Add or remove a track or a curve. Type a minimum and a maximum, or leave them empty for auto scale. Select **log** for a log scale.
3. Tap **Default tracks** to go back to the default set. Tap **Done**.

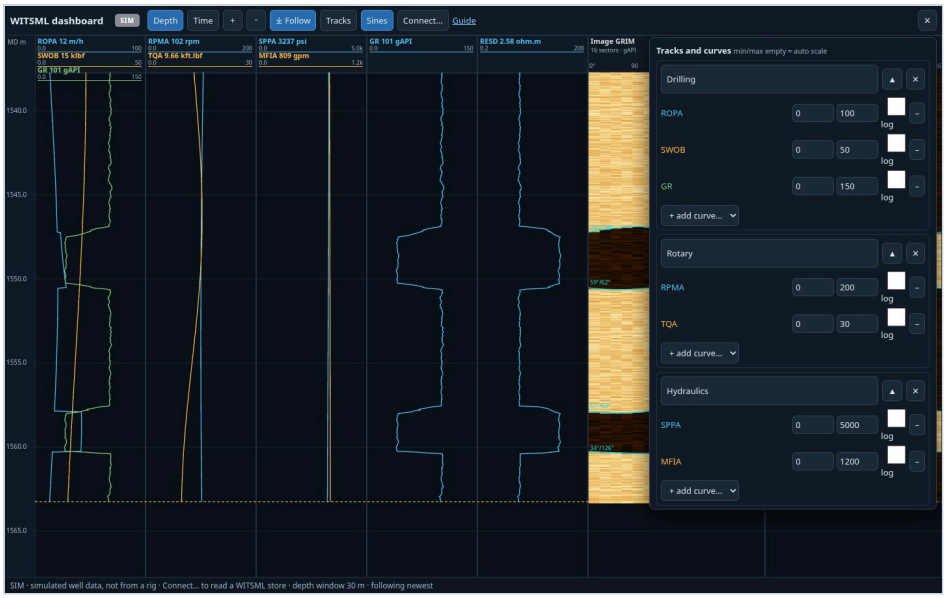



Figure 19a. Laptop (1440 × 900). Track and curve picker.

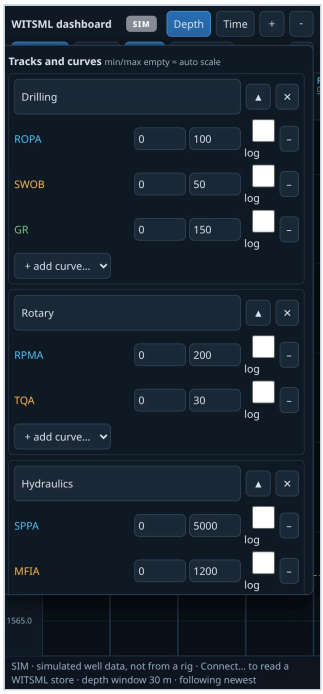



Figure 19b. Phone (390 × 844).

## Connect to a WITSML store

1. Tap **Connect...**.
2. Select **WITSML 1.4.1.1 (SOAP)**.
3. Select the route: **Through a proxy** or **Direct from browser**.
4. Type the store URL, user name and password. Tap **Connect**.
5. Select the well. The dashboard reads all of the well: all wellbores, all time and depth logs (including formation evaluation and mud logs), trajectories (surveys) and mud logs.
6. Tap **Load everything in this well** to read the well again. The panel shows the counts it found.

## Select a dataset

1. The dashboard selects the live (realtime) branch automatically. If there is no realtime log, it selects the newest time log, then the deepest depth log.
2. To show a different dataset, use the dataset list in the top bar. The live branch is ● **LIVE branch**.
3. Each dataset keeps its own index (time or depth).

## Depth and Time mode

1. Tap **Depth** or **Time** in the top bar. This also applies to the image log views.
2. Channels from the other index are put on the same scale with a time-to-depth map. The map uses a bit depth channel (for example DBTM) or a hole depth channel (for example DMEA) from a time log.
3. The note line shows "index map time ↔ depth via" and the channel name.
4. If the wellbore has no bit depth or hole depth channel, the note shows "no time ↔ depth mapping" and lists the channels that are not available in this mode.

> Only the selected live branch updates every 10 s. The other logs are read once, for the last 12 hours or the last 500 m. Mapped values are interpolated. If 2 datasets have the same channel name, the second one gets a suffix, for example `GR@MWD` .

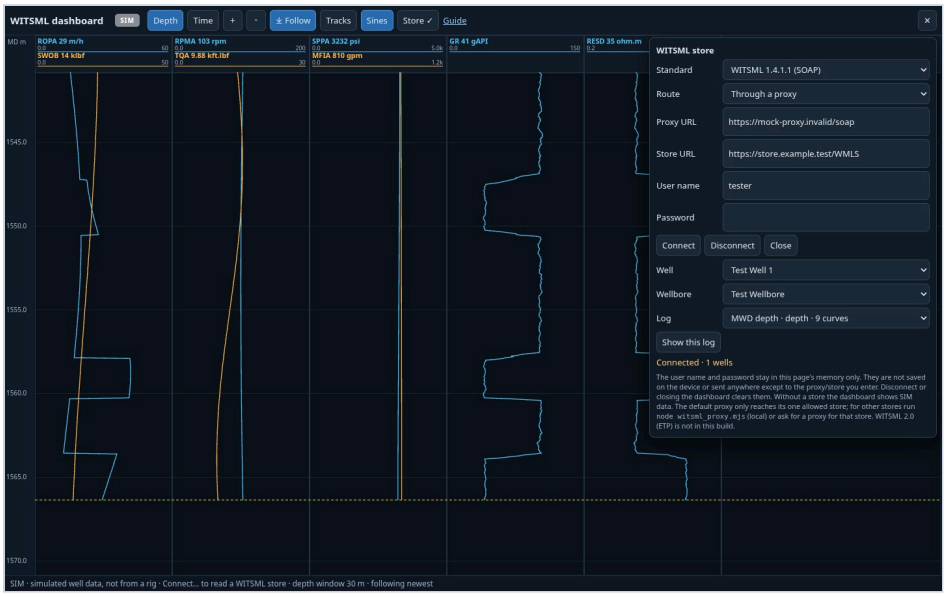



Figure 20a. Laptop (1440 × 900). Connect panel. The values are test values.

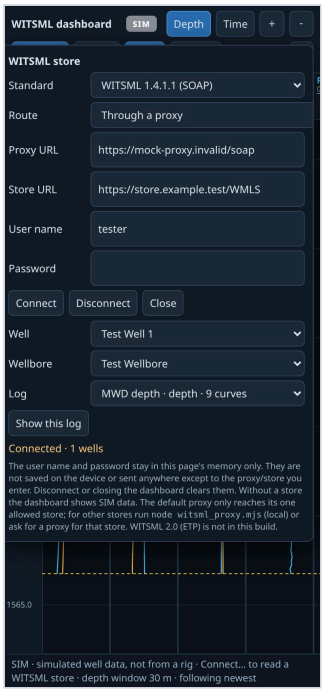



Figure 20b. Phone (390 × 844).

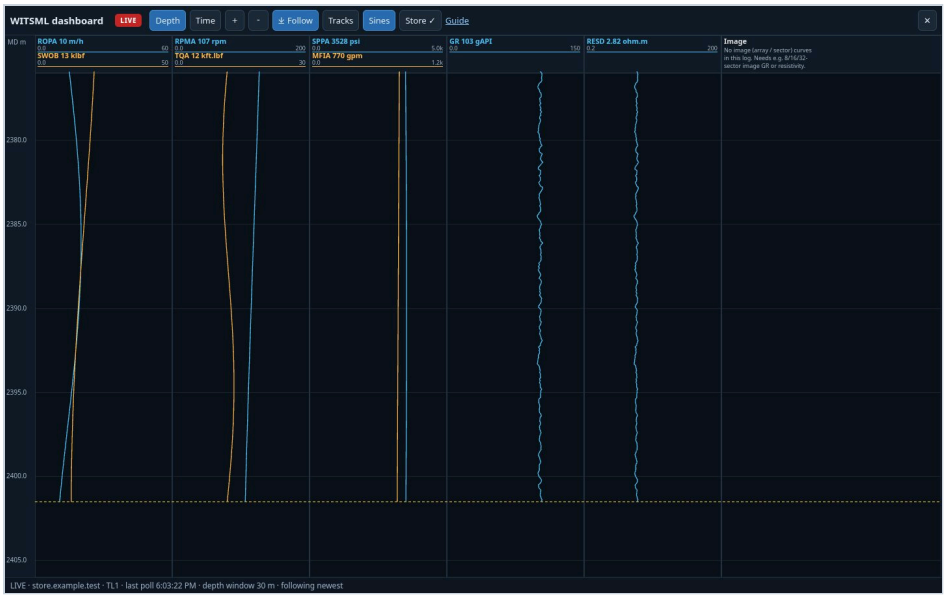



Figure 21a. Laptop (1440 × 900). LIVE tag with a test store.

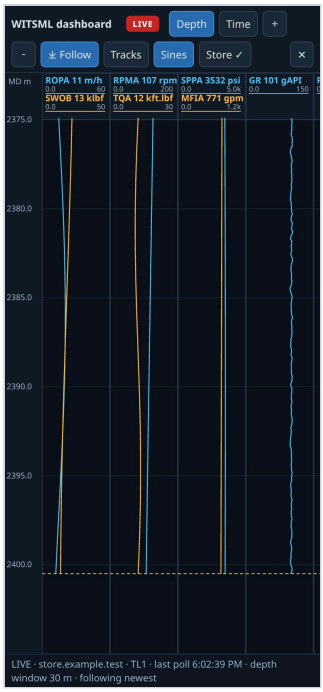



Figure 21b. Phone (390 × 844).

## WITSML in the simulator and Monitor a well

From v5.12.20 the simulator uses the same whole-well model as the dashboard. It reads every wellbore, log, trajectory and mud log in the well. Then it selects the live branch.

1. Connect to WITSML in the simulator and load a well.
2. The **Whole well** box shows what was found. The live branch is selected, and it drives the feed and the 3D wellbore path. If the live branch is on a different wellbore, the simulator changes to that wellbore.
3. To use a different dataset, select it in the branch list. The feed restarts.
4. Tap **Depth** or **Time** to change the index, the same as in the dashboard.
5. In **Monitor a well**, the **Whole well** card shows the same live branch and the Depth/Time buttons.

> Test data is tagged **SIM-TEST**, never LIVE. The 3D view draws only the trajectory of the live branch's wellbore. Credentials stay on your device only.




Figure 21c. Laptop (1440 × 900). Whole well box in the simulator, SIM-TEST data.

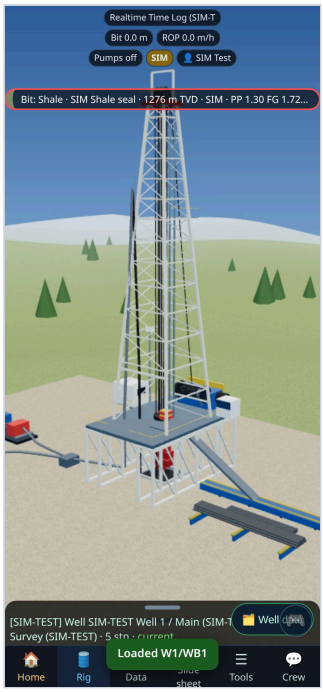



Figure 21d. Phone (390 × 844).




Figure 21e. Laptop (1440 × 900). Whole well card in Monitor a well.

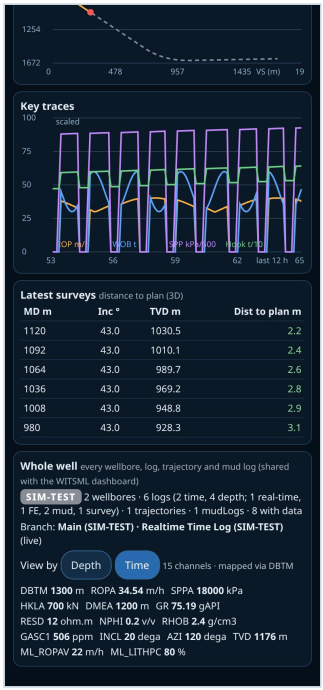



Figure 21f. Phone (390 × 844).

The dashboard remembers your connection on this device only: route, proxy URL, store URL, user name and your last-picked well, wellbore and log. The password is kept until you close the browser tab. Tick **Remember password on this device** to keep it encrypted on this device so the dashboard reconnects by itself when you open it; leave the password blank to reuse the saved one. If the store logs you out or times out, the dashboard retries once with the saved login. The password is never synced, exported or logged. Use **Forget saved connection** on a shared computer to clear everything. If a step fails, the status line says whether it was a login problem, a proxy/CORS problem, an empty result or a store (SOAP) error.

> A browser can read a store directly only if the store allows CORS. If it does not, use a proxy. The default proxy reaches one allowed store only. For a different store, run `node witsml_proxy.mjs` on your computer and use `http://localhost:8796/soap`. WITSML 2.0 (ETP) is not in this version.

The tag is **LIVE** only when all these are true: a store is connected, the log type agrees with the mode, and new data came in the last 2 minutes. If not, the tag shows the cause.